ASSIGNMENT REPORT

Digital Design

SYS_TOP — UART-Controlled Register File & ALU System

Course	*Digital Design*
Assignment	*Final System Project — RTL, Verification, Synthesis, DFT, Formality, CDC & Place and Route*
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Submission Date	*October 2026*

01 OBJECTIVE

This report documents the complete design flow of **SYS_TOP**, a UART-controlled digital system that receives command frames from a master (the testbench), executes register-file and ALU operations, and returns the results over UART. It covers the architecture, RTL implementation, self-checking simulation with waveform analysis, logic synthesis, scan (DFT) insertion, formal equivalence checking (Formality), static lint/CDC sign-off (SpyGlass) and physical implementation through place and route (Cadence First Encounter).

Reports and logs from the tool runs are condensed to their summaries; full raw reports are not reproduced.

Results at a glance

Stage	Result
Functional simulation	5 / 5 test cases pass (register-file write and read, ALU add, subtract, multiply)
Clocks	REF_CLK 50 MHz and UART_CLK 3.6864 MHz (asynchronous); UART at 115.2 kbaud
Synthesis	19,918.00 area units, no constraint violations (TSMC 0.13 µm, Design Compiler)
DFT	4 scan chains (91 / 90 / 90 / 90 cells), **99.45 %** test coverage over 16,140 stuck-at faults
Formality	RTL ≡ post-synthesis (358 points) and RTL ≡ post-DFT (365 points), no failures
SpyGlass CDC	0 unsynchronized crossings (7 scalar and 14 vector crossings synchronized)
Place and route	Post-route setup WNS +0.450 ns, hold WNS +0.030 ns; DRC, connectivity and antenna checks clean
Power	0.227 mW after synthesis; 0.939 mW after routing (different tool and activity assumptions)

Table 1: Results at a glance.

Figure 1 shows how the steps connect: the RTL feeds simulation, lint/CDC sign-off and synthesis; the synthesized netlist is checked by Formality, receives scan chains, is checked again, and is then placed and routed.

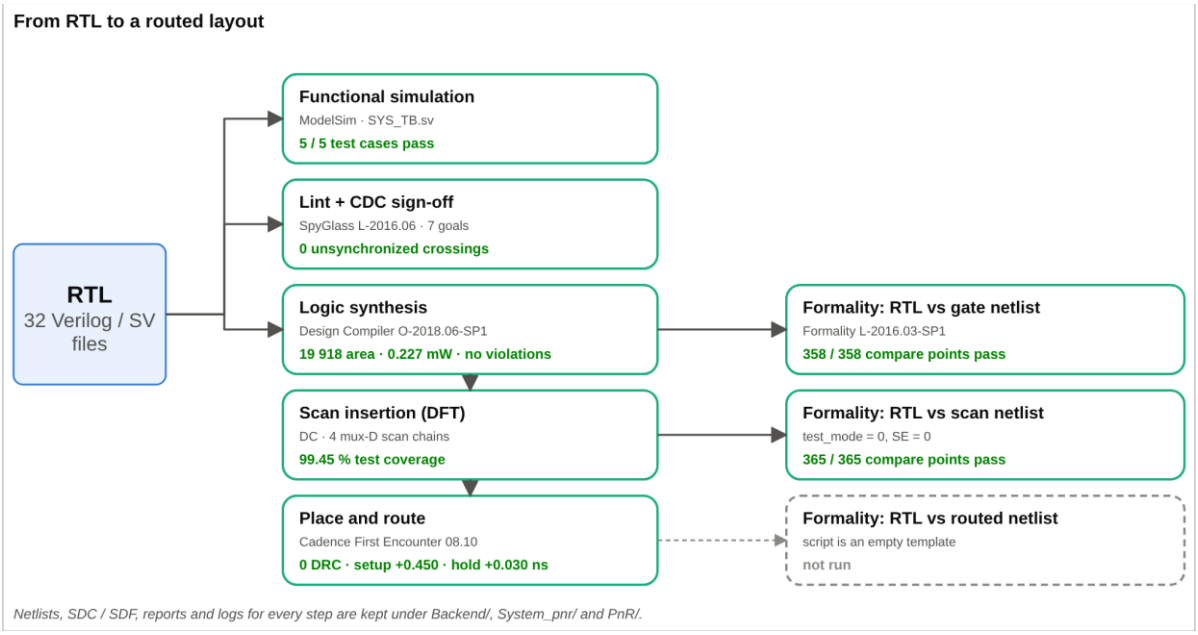


Figure 1: Design and verification flow covered by this report, with the headline result of each step.

02 PROBLEM STATEMENT / REQUIREMENTS

The system must execute commands received serially on `RX_IN` and send any result back on `TX_OUT`. Two asynchronous clock domains are involved, so the design must also handle clock-domain crossing (CDC) and reset synchronization safely.

System specifications

- Reference clock `REF_CLK` = **50 MHz** (period 20 ns); UART clock `UART_CLK` = **3.6864 MHz** (period ≈ 271.3 ns).
- The clock divider is always enabled (`clk_div_en` = 1).
- Top-level ports: inputs `REF_CLK`, `UART_CLK`, `RST` (active-low, asynchronous), `RX_IN`; outputs `TX_OUT`, `RF_PAR_ERR`, `RF_STP_ERR`.
- Ten functional blocks: RegFile, ALU, Clock Gating, SYS_CTRL (REF_CLK domain); UART_TX, UART_RX, PULSE_GEN, Clock Dividers (UART_CLK domain); RST synchronizers, Data synchronizer and ASYNC FIFO (crossings).
- Start-up sequence: configure UART through register-file writes to addresses 0x2 and 0x3, then the master sends RegFile / ALU commands; SYS_CTRL executes them and returns results via UART_TX.
- Addresses 0x0–0x3 are reserved for ALU operands and configuration; the spec describes the remaining range for normal read/write operations as starting at 0x4.

Supported commands

Command	Opcode	Frame sequence
RF Write (3 frames)	0xAA	Frame 0 = 0xAA, Frame 1 = address, Frame 2 = write data
RF Read (2 frames)	0xBB	Frame 0 = 0xBB, Frame 1 = address → 1 byte returned
ALU with operands (4 frames)	0xCC	Frame 0 = 0xCC, Frame 1 = operand A, Frame 2 = operand B, Frame 3 = ALU_FUN → 2 bytes returned (LSB, MSB)
ALU without operands (2 frames)	0xDD	Frame 0 = 0xDD, Frame 1 = ALU_FUN (uses the A/B already in REG0/REG1) → 2 bytes returned

Table 2: Command set. Each UART frame = 1 start bit, 8 data bits (LSB first), parity bit, 1 stop bit (11 bits).

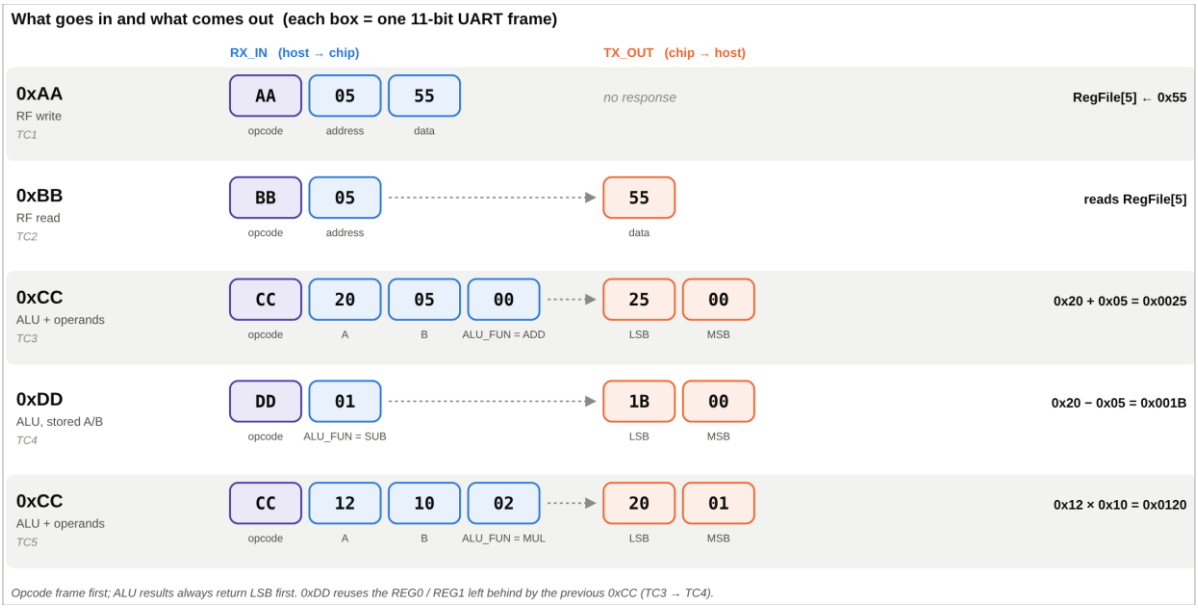


Figure 2: Byte sequences on RX_IN and the bytes returned on TX_OUT for each command.

Configuration registers

Register	Address	Function	Reset value
REG0	0x0	ALU operand A	0x00
REG1	0x1	ALU operand B	0x00
REG2	0x2	UART config: [0] parity enable, [1] parity type (0 = even), [7:2] prescale	0x81 (parity on, even, prescale 32)
REG3	0x3	Clock-divider ratio for the TX clock	0x20 (÷32)

Table 3: Reserved register-file locations (reset values taken from the RegFile RTL).

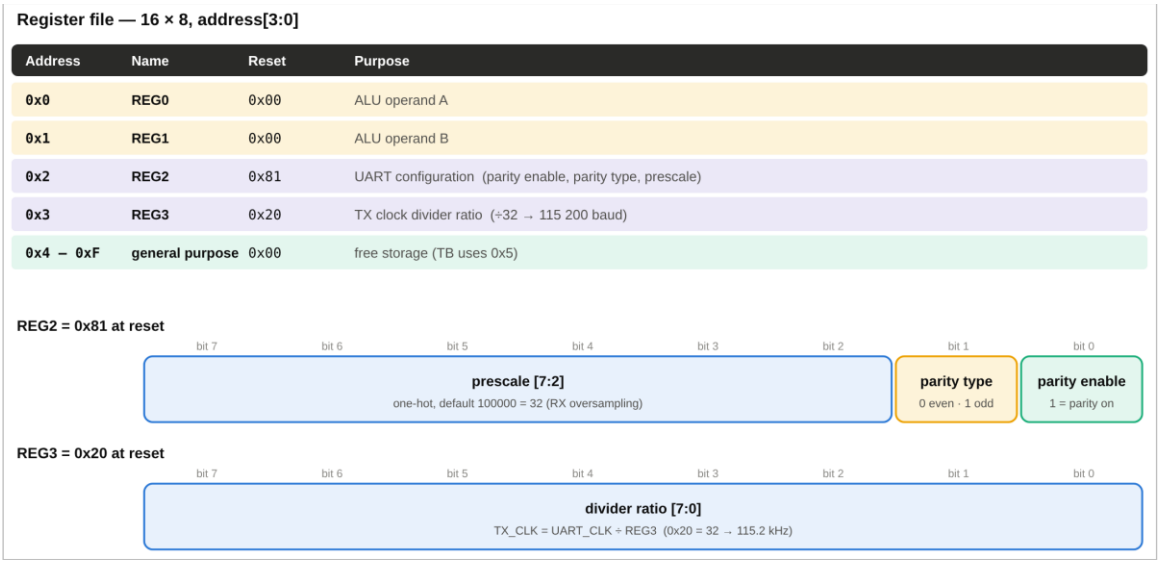


Figure 3: Register-file map, with the reserved locations REG0–REG3 and the REG2 / REG3 bit fields.

03 DESIGN & ANALYSIS

3.1 Design Approach

The design is partitioned by clock domain, with every crossing handled by a dedicated synchronizer structure:

- **Reset:** one 2-flop reset synchronizer per domain (`RST_SYNC_1` on REF_CLK, `RST_SYNC_2` on UART_CLK) gives asynchronous assertion and synchronous de-assertion.
- **RX path (UART_CLK → REF_CLK):** the received byte and its valid flag cross through `DATA_SYNC` (2-flop synchronizer on the enable with edge-detect pulse generation that qualifies the bus).
- **TX path (REF_CLK → UART_CLK):** SYS_CTRL pushes result bytes into an 8-entry **asynchronous FIFO** with Gray-coded pointers and 2-flop pointer synchronizers. `PULSE_GEN` converts the UART_TX `busy` level into a one-cycle `R_INC` read pulse.
- **Static configuration:** REG2 (parity/prescale) and REG3 (divider ratio) are written once at start-up and then held, so they connect directly into the UART domain as quasi-static signals.
- **Baud-rate generation:** the TX divider divides UART_CLK by REG3 (32), giving 3.6864 MHz / 32 = **115.2 kHz** (bit period 8.68 µs). The RX divider ratio is decoded from the one-hot prescale field by `CLKDIV_MUX` (32→1, 16→2, 8→4, 4→8) so RX always samples at a fixed multiple of the baud rate; RX uses a 3-sample majority vote per bit.
- **Power:** the ALU clock is gated by a latch-based clock gate. SYS_CTRL enables it only while an ALU operation is in progress and removes it once `OUT_VALID` is seen.

Clock and reset structure

Figure 4 shows how the two input clocks and the reset are distributed. `ALU_CLK` is `REF_CLK` gated by `CLK_GATE`; `TX_CLK` is `UART_CLK` divided by REG3; `RX_CLK` is `UART_CLK` divided by the ratio that `CLKDIV_MUX` decodes from the REG2 prescale field; and each domain has its own reset synchronizer.

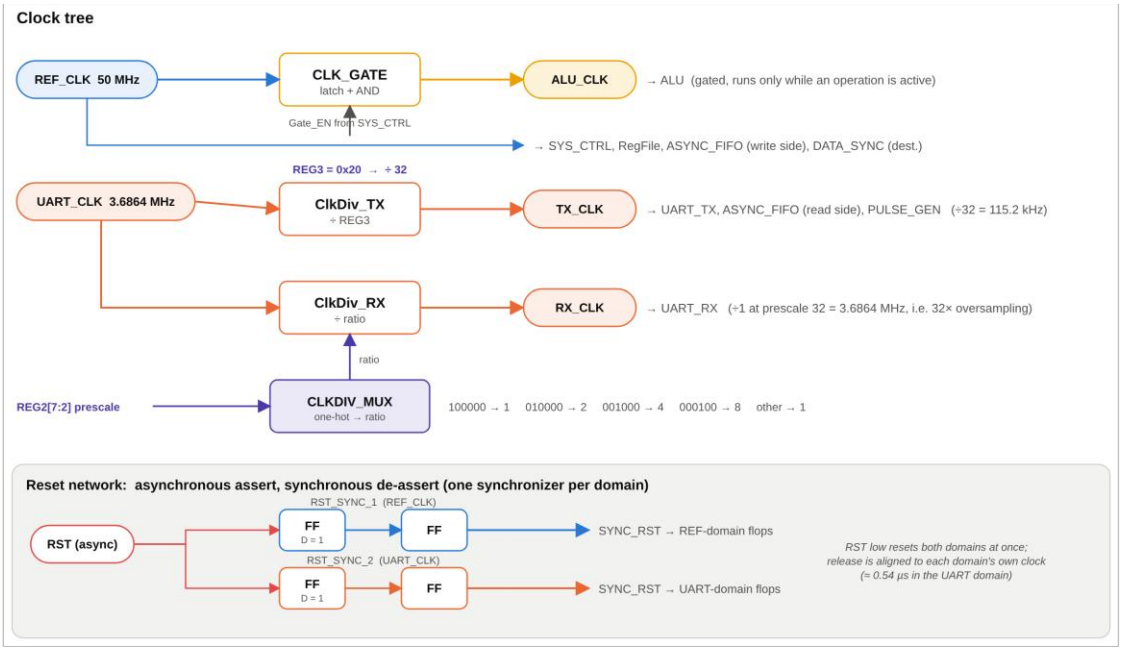


Figure 4: Clock tree and reset network of SYS_TOP.

Asynchronous FIFO. Figure 5 shows the TX-path FIFO. SYS_CTRL writes the 8 × 8 memory in the REF_CLK domain and UART_TX reads it in the TX_CLK domain; Gray-coded read and write pointers cross through 2-flop synchronizers to produce the FULL and EMPTY flags.

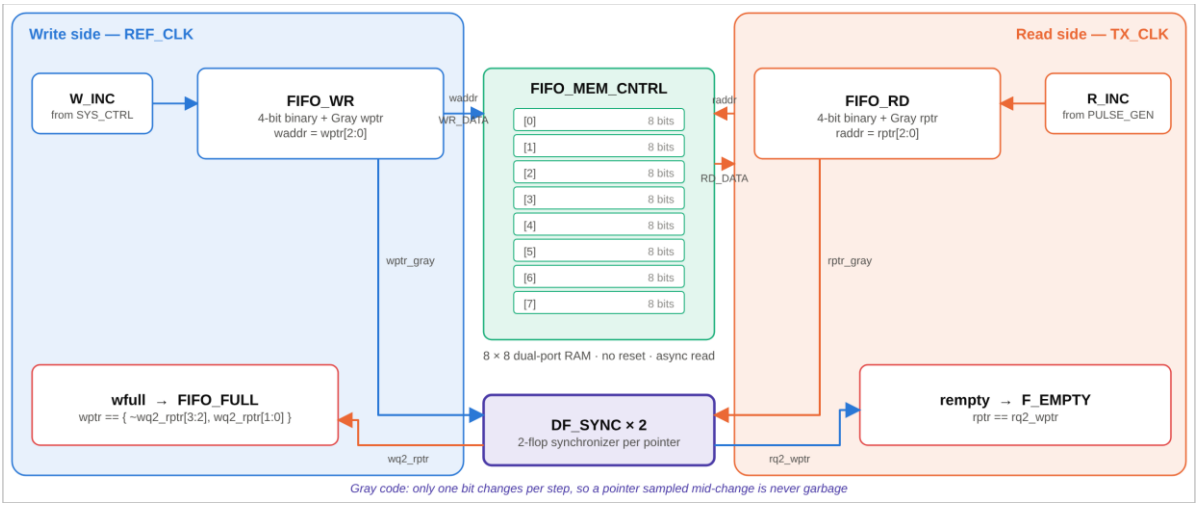


Figure 5: Asynchronous FIFO with Gray-coded pointers and 2-flop pointer synchronizers.

UART frame. Figure 6 shows the 11-bit frame used by UART_TX and UART_RX: a start bit, eight data bits sent LSB first, an even parity bit and a stop bit. UART_RX samples each bit with a 3-sample majority vote around the middle of the bit period.

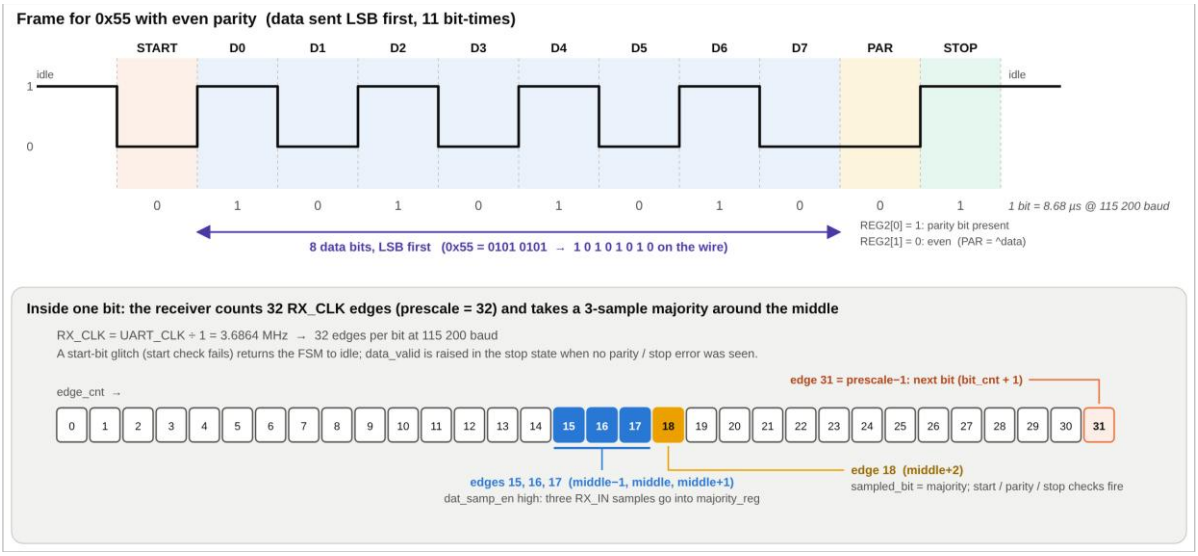


Figure 6: UART frame format and receiver sampling.

3.2 State Table

SYS_CTRL is a 14-state FSM clocked by REF_CLK. The first received byte selects the command; later frames are captured in `frame1/2/3_reg`. Control outputs are one-cycle pulses.

State	Code	Behaviour
S_IDLE	0	Wait for `RX_D_VLD`; latch opcode into `cmd_reg`
S_FRAME1	1	Capture frame 1; RF-Read → S_RF_READ_REQ, ALU-NOP → S_ALU_EXEC, else S_FRAME2; unknown opcode → IDLE
S_FRAME2	2	Capture frame 2; RF-Write → S_RF_WRITE, ALU-OP → S_FRAME3
S_FRAME3	3	Capture ALU_FUN frame → S_ALU_WR_A
S_RF_WRITE	4	Pulse `WrEn` with `Address = frame1[3:0]`, `WrData = frame2`
S_RF_READ_REQ	5	Pulse `RdEn` with the requested address
S_RF_READ_WAIT	6	Wait for `RdData_Valid`
S_RF_READ_SEND	7	If FIFO not full: push `RdData` to FIFO (`TX_D_VLD`)
S_ALU_WR_A / WR_B	8 / 9	Write operands into REG0 (addr 0) and REG1 (addr 1); enable the gated clock
S_ALU_EXEC	10	Assert `EN`, drive `ALU_FUN` (frame 3 for 0xCC, frame 1 for 0xDD)
S_ALU_WAIT	11	Wait for `OUT_Valid`, then drop `CLK_EN`
S_ALU_SEND_LSB / MSB	12 / 13	Push `ALU_OUT[7:0]` then `ALU_OUT[15:8]` to the FIFO when not full

Table 4: SYS_CTRL state table (from the RTL).

Figure 7 traces the path SYS_CTRL takes through the state table for each of the four commands.

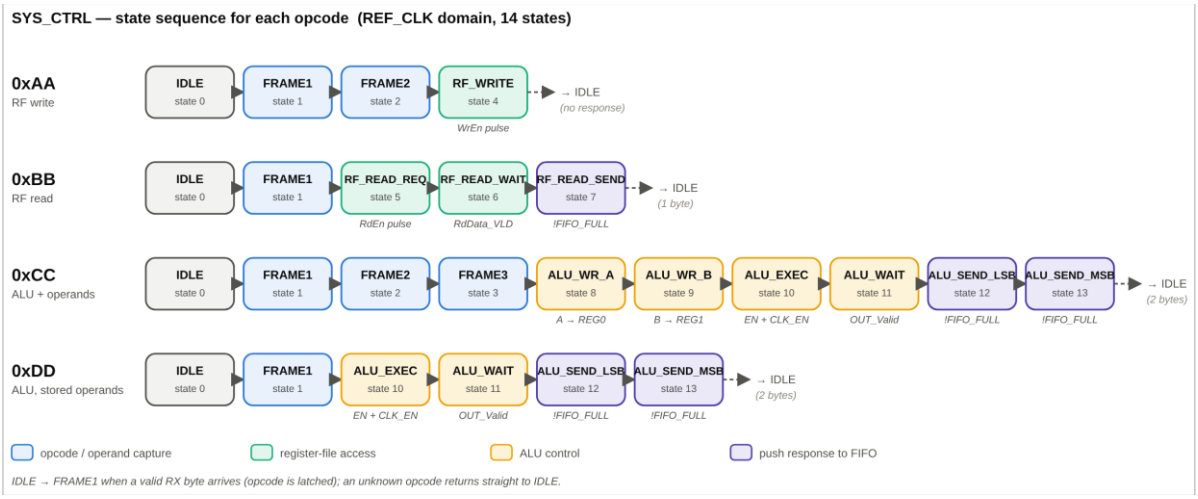


Figure 7: SYS_CTRL state paths for each command.

The ALU is registered, 16-bit wide at its output, and selected by a 4-bit ALU_FUN:

FUN	Operation	FUN	Operation	FUN	Operation
0000	A + B	0110	~(A & B)	1100	A < B → 3
0001	A − B	0111	~(A | B)	1101	A >> 1
0010	A × B	1000	A ^ B	1110	A << 1
0011	A ÷ B	1001	A ~^ B	—	—
0100	A & B	1010	A == B → 1	—	—
0101	A | B	1011	A > B → 2	—	—

Table 5: ALU function encoding (from ALU.v). The A < B comparison is implemented in addition to the operations listed in the spec.

3.3 Circuit / Block Diagram

Figure 8 shows the system architecture from the specification and Figure 9 shows the SYS_TOP symbol as seen after synthesis. Figure 10 redraws the architecture as a datapath with the clock-domain crossings highlighted.

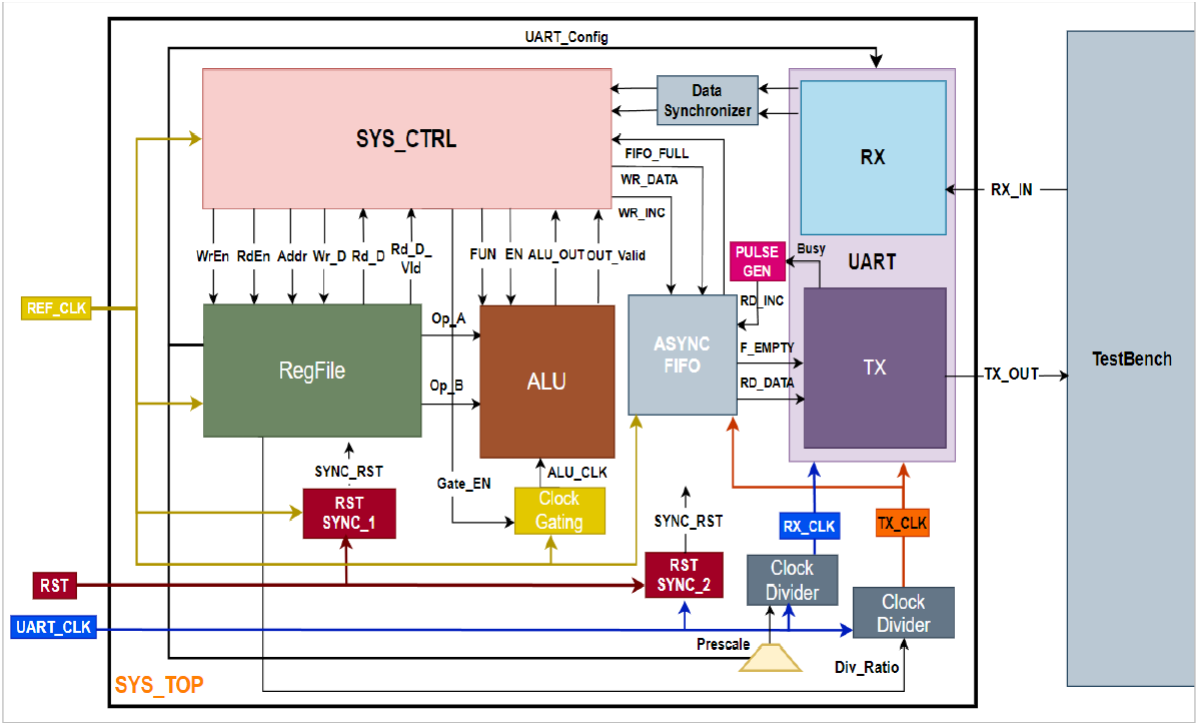


Figure 8: SYS_TOP block diagram — two clock domains, reset synchronizers, data synchronizer and asynchronous FIFO.



Figure 9: SYS_TOP top-level interface (REF_CLK, UART_CLK, RST, RX_IN → TX_OUT, RF_PAR_ERR, RF_STP_ERR).

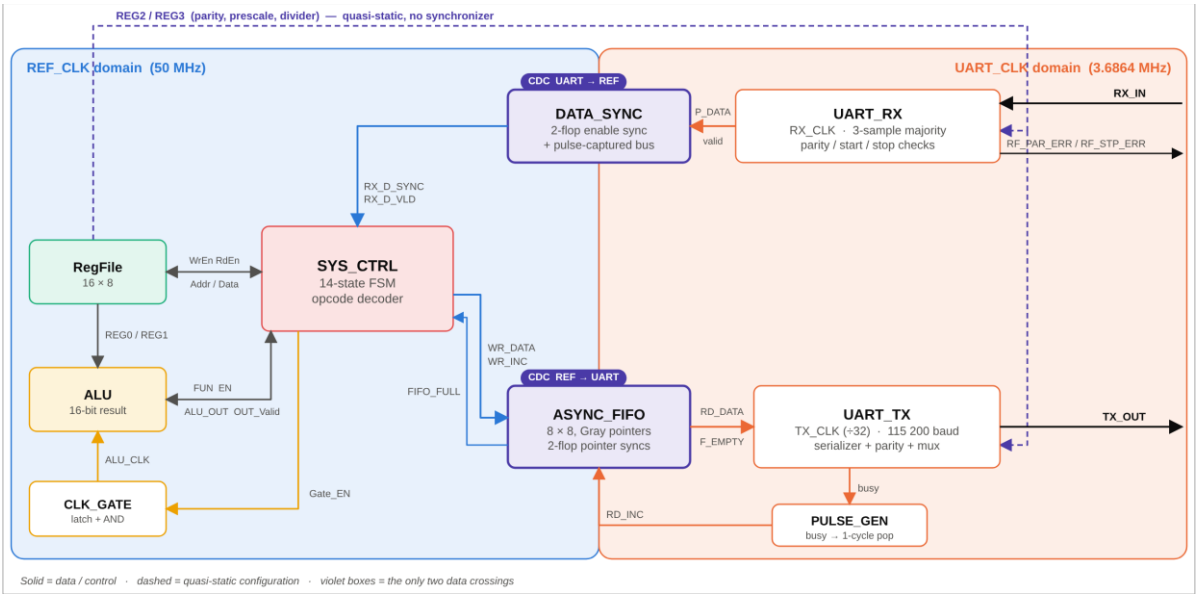


Figure 10: Datapath view of SYS_TOP with the clock-domain crossings.

Block	Domain	Role
RegFile	REF_CLK	16 × 8-bit register file; REG0–REG3 exposed as dedicated outputs
ALU	ALU_CLK (gated REF_CLK)	15 arithmetic/logic/compare/shift functions, registered 16-bit result + valid
CLK_GATE	REF_CLK	Latch + AND clock gate controlled by SYS_CTRL `CLK_EN`
SYS_CTRL	REF_CLK	Command decoder / FSM linking UART, RegFile, ALU and FIFO
UART_RX / UART_TX	RX_CLK / TX_CLK	Serial receiver (majority-vote sampling, parity & stop checks) and transmitter (serializer, parity calculation)
ClkDiv ×2, CLKDIV_MUX	UART_CLK	TX divider (REG3) and RX divider (decoded from prescale)
PULSE_GEN	TX_CLK	Rising-edge detector turning UART `busy` into FIFO read increment
RST_SYNC ×2, DATA_SYNC, ASYNC_FIFO	Crossings	Reset synchronization, RX bus crossing, TX data crossing

Table 6: Block summary.

3.4 HDL Implementation

The design is written in Verilog with SystemVerilog FSMs for the UART blocks (about 30 source files). The top level `SYS_TOP.v` instantiates every block and is parameterized (`RF_WIDTH=8`, `RF_DEPTH=16`, `RF_ADDR=4`, `FIFO_WIDTH=8`). Key excerpts follow.

TX crossing: FIFO + pulse generator (SYS_TOP.v)

```
PULSE_GEN U_PULSE_GEN (
    .clk(TX_CLK), .rst(SYNC_RST_2),
    .lvl_sig(UART_TX_BUSY), .pulse_sig(FIFO_R_INC)
);

ASYNC_FIFO #(.DATA_WIDTH(FIFO_WIDTH)) U_ASYNC_FIFO (
    .W_CLK(REF_CLK), .W_RST(SYNC_RST_1), .W_INC(SYS_TX_D_VLD),
    .R_CLK(TX_CLK),  .R_RST(SYNC_RST_2), .R_INC(FIFO_R_INC),
    .WR_DATA(SYS_TX_P_DATA), .FULL(FIFO_FULL),
    .EMPTY(FIFO_EMPTY),      .RD_DATA(FIFO_RD_DATA)
);
```

Clock gate behavioural model (CLK_GATE.v)

```
always @(CLK or CLK_EN)
  if (!CLK) Latch_Out <= CLK_EN;      // transparent while CLK is low
assign GATED_CLK = CLK && Latch_Out;  // glitch-free gated clock
```

For synthesis, this model is replaced by the library integrated clock-gating cell (`TLATNCAX12M`), which appears as a latch compare point in the Formality results.

Pulse generator (PULSE_GEN.v)

```
always @(posedge clk or negedge rst)
  if (!rst) begin rcv_flop <= 0; pls_flop <= 0; end
  else      begin rcv_flop <= lvl_sig; pls_flop <= rcv_flop; end
assign pulse_sig = rcv_flop && !pls_flop;   // one-cycle pulse on rising edge
```

04 SIMULATION & RESULTS

4.1 Verification Approach

Verification uses a self-checking SystemVerilog testbench (`SYS_TOP_tb`) run in ModelSim. The testbench acts as the UART master and also monitors `TX_OUT`:

- **Clocks:** REF_CLK period 20 ns; UART_CLK period 271.267 ns; UART bit period 8680.55 ns (115.2 kbaud).
- **Reset:** `RST` held low for 200 ns then released, followed by 1 µs of wait so the reset synchronizers de-assert before traffic starts.
- **Configuration over UART:** writes REG2 = 0x81 (parity enabled, even, prescale 32) and REG3 = 0x20 (divide by 32), as required by the spec sequence.
- **Driver:** `send_uart_frame` generates start bit, 8 data bits LSB-first, even parity (`^data`) and stop bit.
- **Monitor:** on each test trigger it waits for the falling edge of `TX_OUT` (with a 200-bit-period timeout), samples every bit mid-cell, and compares each received byte with the expected value. For the RegFile write case no TX data is expected, so it waits 50 bit periods and prints `REGFILE[5]`.
- **Scoreboard:** counts total/passed/failed cases and prints a summary.

4.2 Test Results

TC	Test	Frames sent	Expected TX	Observed	Result
1	RegFile write 0x55 → addr 0x05	0xAA, 0x05, 0x55	none (RF[5] = 0x55)	RF[5] = 0x55	PASS
2	RegFile read addr 0x05	0xBB, 0x05	0x55	0x55	PASS
3	ALU ADD with operands 0x20 + 0x05	0xCC, 0x20, 0x05, 0x00	0x25, 0x00	0x25, 0x00	PASS
4	ALU SUB, no operands (0x20 – 0x05)	0xDD, 0x01	0x1B, 0x00	0x1B, 0x00	PASS
5	ALU MUL with operands 0x12 × 0x10	0xCC, 0x12, 0x10, 0x02	0x20, 0x01	0x20, 0x01	PASS

Table 7: Test cases and results. TC4 reuses the operands left in REG0/REG1 by TC3. Results are returned LSB first (0x0025, 0x001B, 0x0120).

```
VSIM 21> run -all
#
# === STARTING TESTBENCH ===
#
# --- TC#1: RegFile Write [0x55 -> Addr 0x05] ---
#   [REGFILE[5]] = 0x55
# TC#1: PASSED
# --- TC#2: RegFile Read [Addr 0x05, Exp: 0x55] ---
#   [MATCH] Byte 0: 0x55
#   [REGFILE[5]] = 0x55
# TC#2: PASSED
# --- TC#3: ALU ADD [0x20 + 0x05 -> 0x0025] ---
#   [MATCH] Byte 0: 0x25
#   [MATCH] Byte 1: 0x0
#   [REGFILE[5]] = 0x55
# TC#3: PASSED
# --- TC#4: ALU SUB [0x20 - 0x05 -> 0x001B] ---
#   [MATCH] Byte 0: 0x1b
#   [MATCH] Byte 1: 0x0
#   [REGFILE[5]] = 0x55
# TC#4: PASSED
# --- TC#5: ALU MUL [0x12 * 0x10 -> 0x0120] ---
#   [MATCH] Byte 0: 0x20
#   [MATCH] Byte 1: 0x1
#   [REGFILE[5]] = 0x55
# TC#5: PASSED
#
# === TESTBENCH SUMMARY ===
# Total: 5 | Passed: 5 | Failed: 0
# STATUS: ALL TESTS PASSED!
```

Figure 11: ModelSim transcript — Total 5, Passed 5, Failed 0, "ALL TESTS PASSED".

4.3 Waveform Analysis

How to read the waveforms

All five waveforms use the same signal groups. The time axis is in nanoseconds; one UART frame (11 bits) lasts ≈ 95.5 µs, which is why activity is spread over hundreds of microseconds while the REF_CLK-domain pulses are only 20 ns wide.

Signal / group	What it shows
`U_SYS_CTRL/state`, `next_state`	FSM state. It steps IDLE (0) → FRAME1 (1) → FRAME2 (2) → FRAME3 (3) as frames arrive. Single-cycle states (e.g. S_RF_WRITE, ALU states) last 20 ns and are too narrow to resolve at this zoom.
`test_case_id`, `test_label`, `total_tc`, `passed_tc`, `failed_tc`	Testbench bookkeeping; `passed_tc` increments when a case finishes, `failed_tc` stays 0 throughout.
`tc_start_time`, `expected_num_bytes`, `expected_bytes`	Start time of the case and the bytes the monitor expects on TX.
`U0_UART_RX/P_DATA`, `data_valid`	Byte decoded by UART_RX (hexagons) and its one-cycle valid pulse — one pulse per received frame.
`U0_UART_TX/txOut`, `busy`	Serial transmit line (idle high, start bit low) and the TX busy flag, high for the whole frame.
`U_ASYNC_FIFO/RAM`, `WR_DATA`, `EMPTY`, `FULL`, `RD_DATA`	FIFO storage, the byte pushed by SYS_CTRL (non-zero for one REF_CLK cycle), the empty flag (low while data waits) and the byte read by UART_TX. `FULL` stays low.
`U_RegFile/regArr[0..15]`	Register contents; [2] = 0x81 and [3] = 0x20 confirm the UART configuration.

Table 8: Waveform signal legend.

TC1 — RegFile write (0x55 → address 0x05)

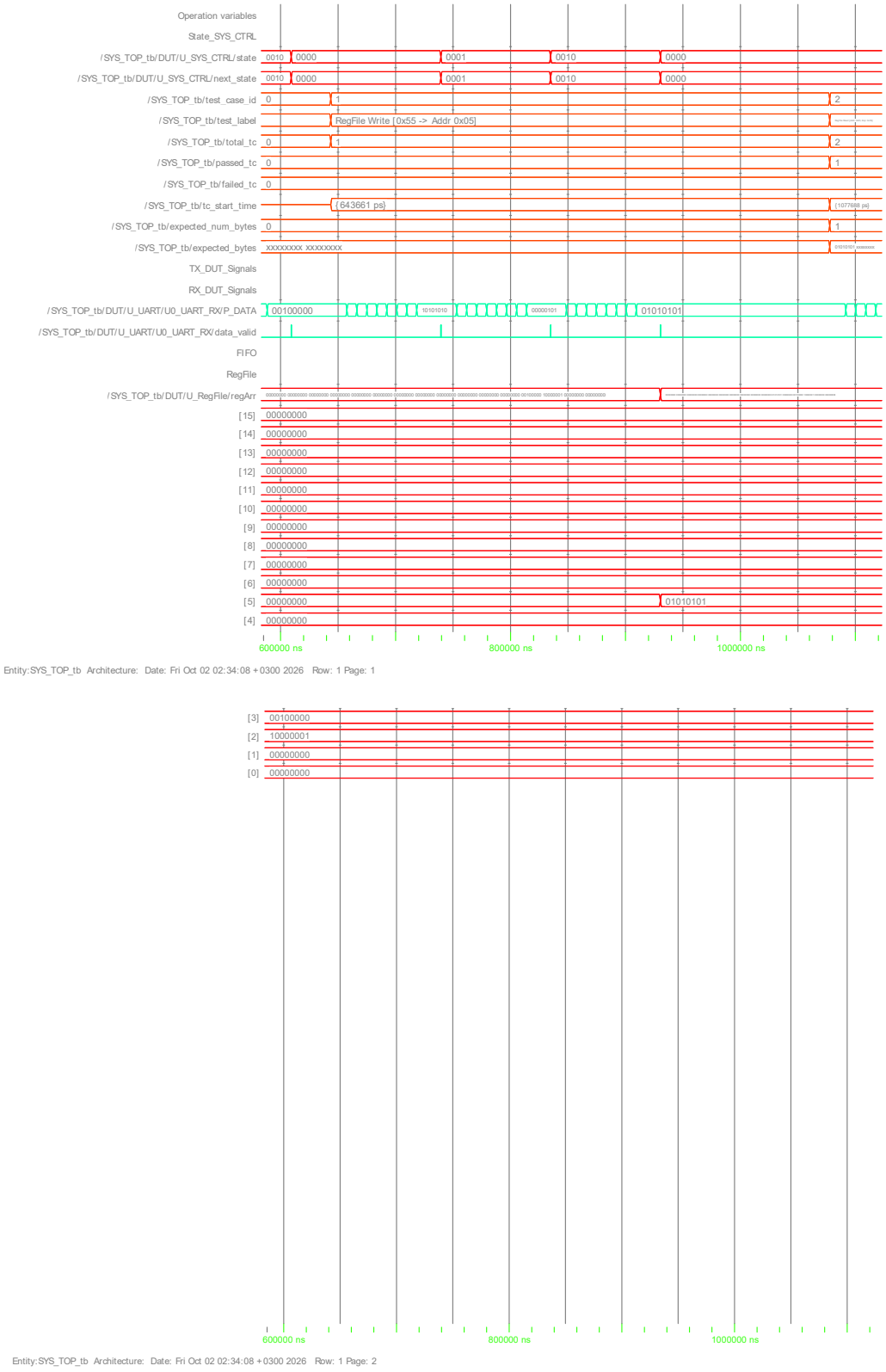
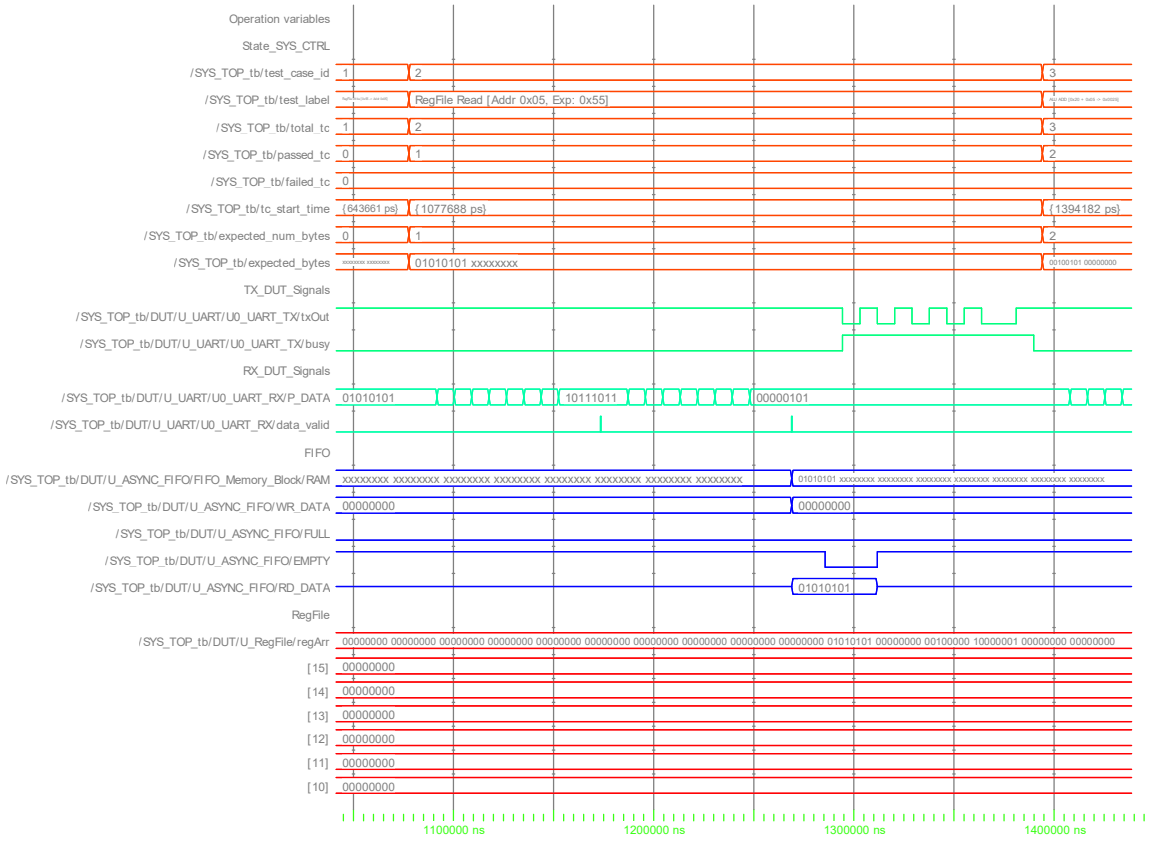


Figure 12: TC1 waveform, ≈ 0.60–1.10 ms.

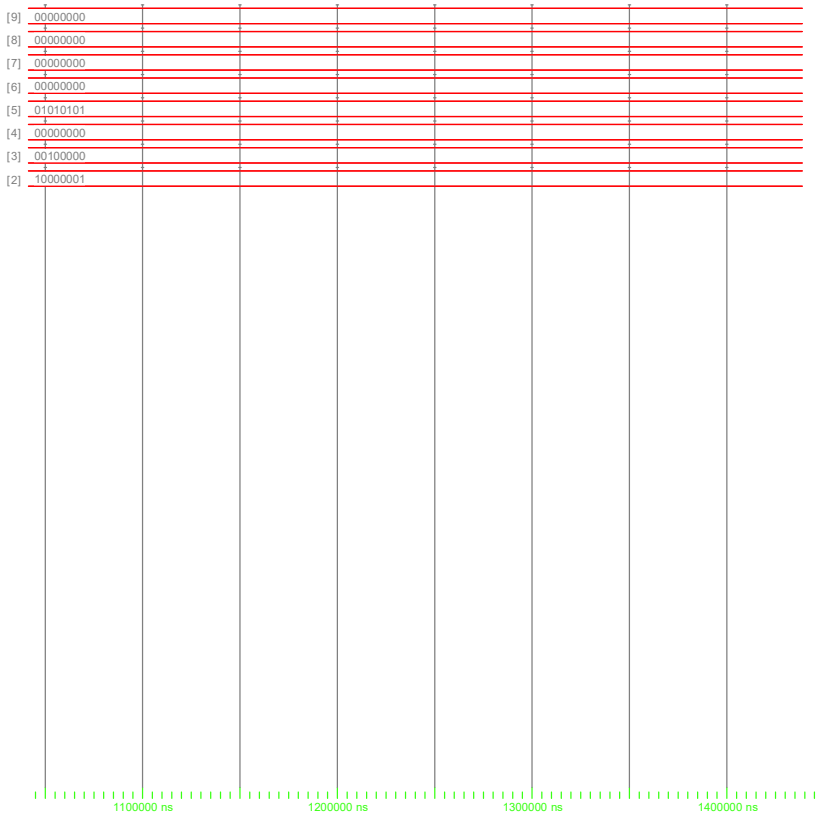
The case starts at ≈ 643.7 µs. UART_RX decodes three frames — 0xAA (`10101010`), 0x05 (`00000101`) and 0x55 (`01010101`) — each marked by a `data_valid` pulse. SYS_CTRL moves 0000 → 0001 → 0010 as frames 0 and 1 arrive, returns to 0000 after the third frame, and `regArr[5]` changes from 0x00 to **0x55**

at ≈ 0.93 ms. The leftmost frame (`00100000`) is the data byte of the earlier REG3 configuration write. No TX activity occurs, so the monitor waits 50 bit periods and the case ends at ≈ 1.078 ms (`passed_tc` = 1).

TC2 — RegFile read (address 0x05, expect 0x55)



Entity:SYS_TOP_tb Architecture: Date: Fri Oct 02 03:08:20 +0300 2026 Row: 1 Page: 1



Entity:SYS_TOP_tb Architecture: Date: Fri Oct 02 03:08:20 +0300 2026 Row: 1 Page: 2

Figure 13: TC2 waveform, ≈ 1.08–1.45 ms.

Two frames arrive: 0xBB (`10111011`) and the address 0x05. SYS_CTRL reads the register and pushes **0x55** into the FIFO — `RAM` shows `01010101`. `EMPTY` goes low, UART_TX reads the byte (`RD_DATA` = `01010101`), `busy` rises and `txOut` sends one frame: start bit low, eight data bits, parity and stop. `busy` falls at the end of the frame, matching the expected byte 0x55.

TC3 — ALU ADD with operands (0x20 + 0x05)

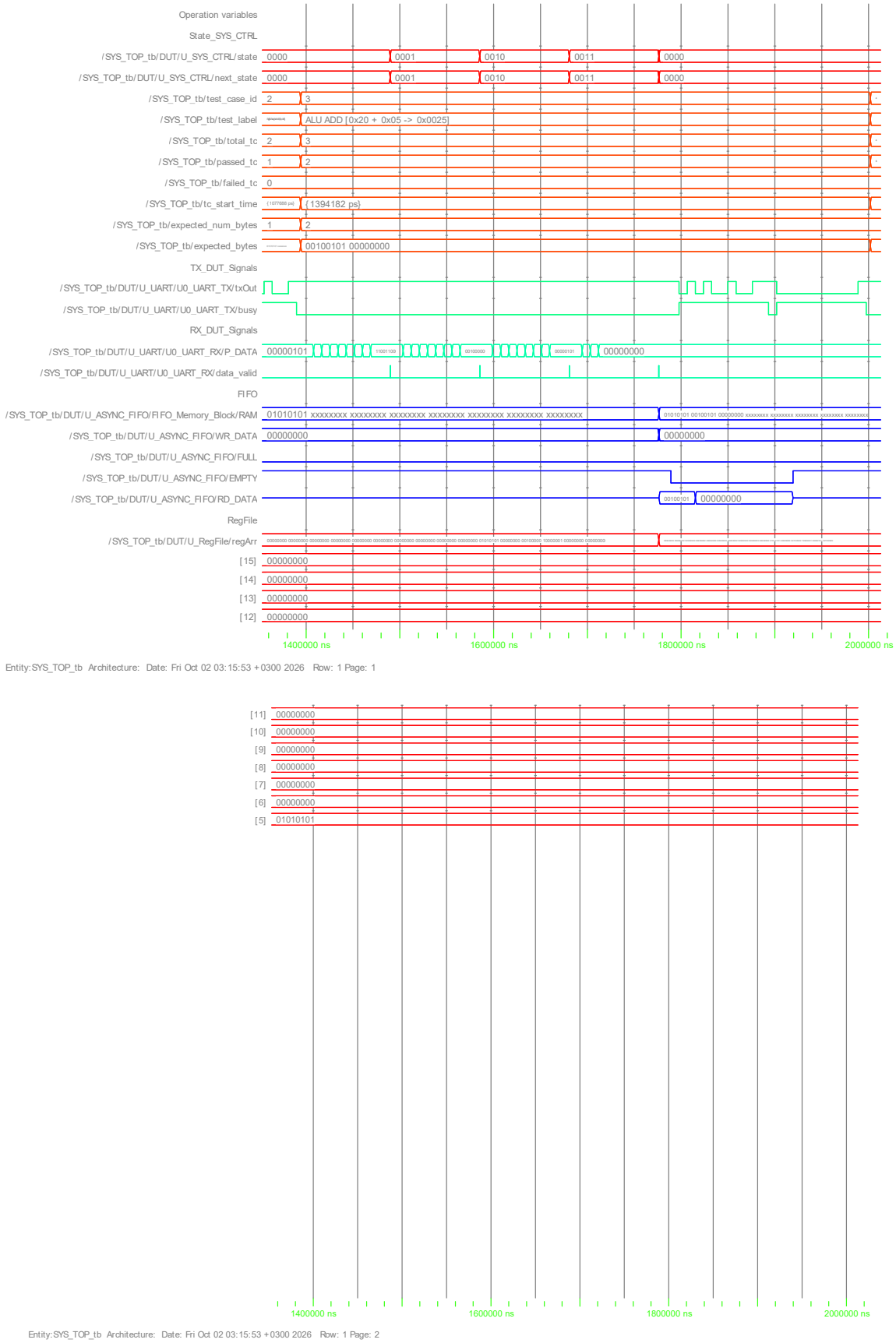


Figure 14: TC3 waveform, ≈ 1.39–2.00 ms.

Four frames are decoded: 0xCC (`11001100`), 0x20 (`00100000`), 0x05 (`00000101`) and the function code 0x00 (ADD). The state trace shows 0000 → 0001 → 0010 → 0011 → 0000; the ALU states (write A, write B, execute, wait, send) complete within a few hundred nanoseconds after the fourth frame and are too narrow to resolve here. The FIFO receives two bytes — `00100101` (0x25) then `00000000` (0x00) — and `txOut` transmits two back-to-back frames while `EMPTY` stays low. The 16-bit result 0x0025 matches the expected value, and `regArr[0]`/`regArr[1]` now hold 0x20 and 0x05.

TC4 — ALU SUB without operands (0x20 − 0x05)

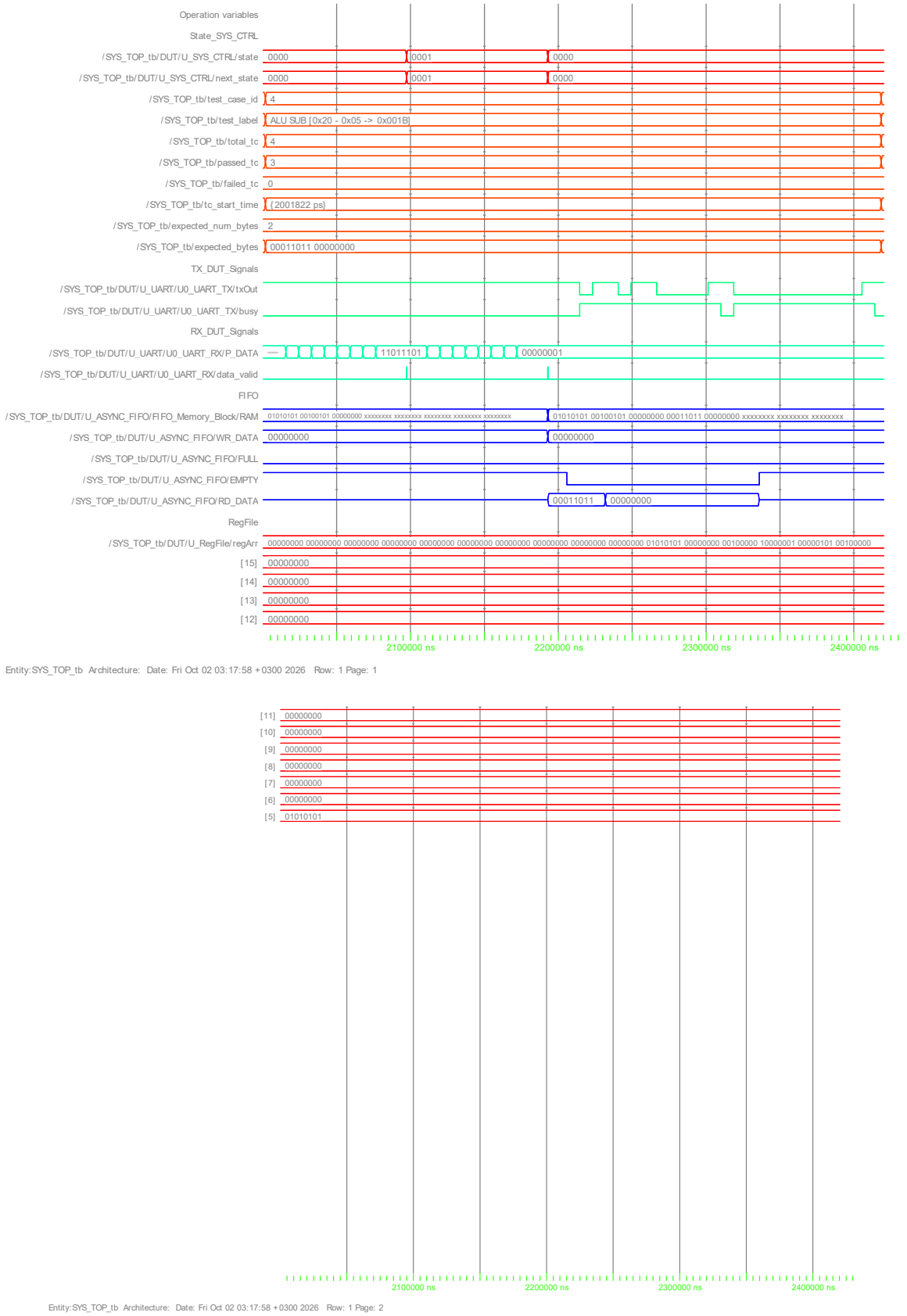
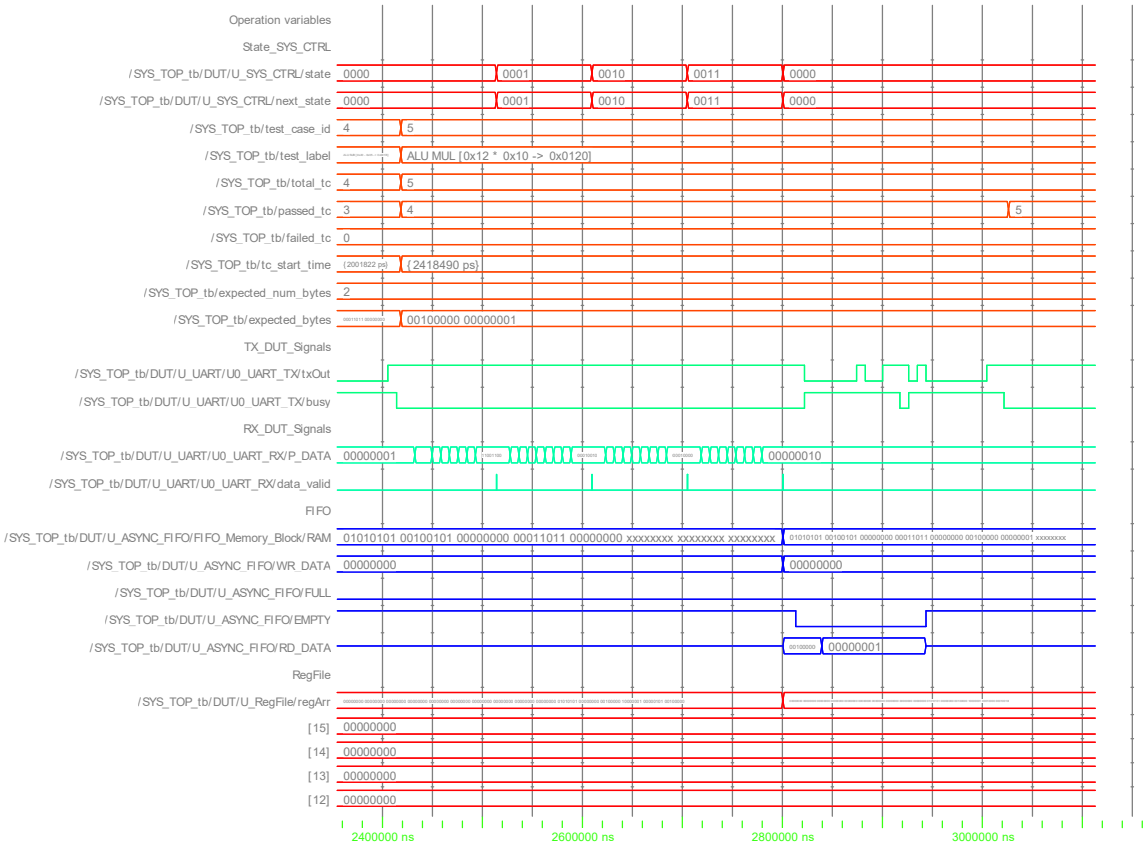


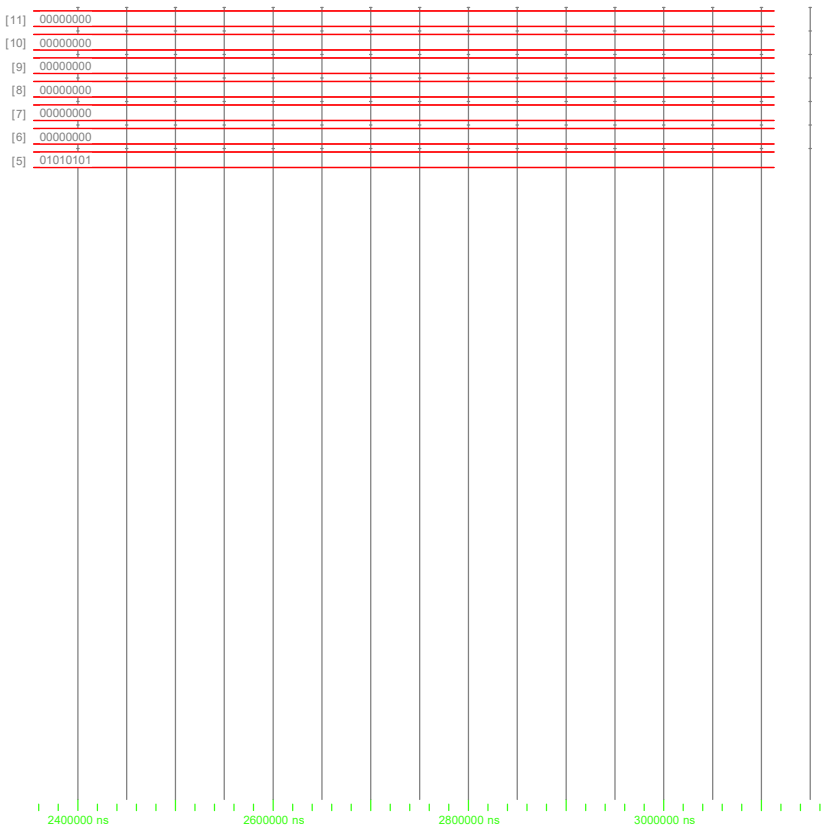
Figure 15: TC4 waveform, ≈ 2.00–2.45 ms.

Only two frames are sent: 0xDD (`11011101`) and the function code 0x01 (SUB). The FSM goes 0000 → 0001 → 0000 because the no-operand command jumps from FRAME1 straight to execution, reusing REG0 = 0x20 and REG1 = 0x05 left by TC3 (visible in `regArr`). The FIFO holds `00011011` (0x1B) and `00000000`, and both bytes appear on `txOut` — result 0x001B as expected.

TC5 — ALU MUL with operands (0x12 × 0x10)



Entity:SYS_TOP_tb Architecture: Date: Fri Oct 02 03:20:27 +0300 2026 Row: 1 Page: 1



Entity:SYS_TOP_tb Architecture: Date: Fri Oct 02 03:20:27 +0300 2026 Row: 1 Page: 2

Figure 16: TC5 waveform, ≈ 2.40–3.10 ms.

The four frames 0xCC, 0x12, 0x10 and 0x02 (MUL) produce the 16-bit product 0x0120, which exceeds one byte. SYS_CTRL therefore sends the low byte `00100000` (0x20) first and the high byte `00000001` (0x01) second, both visible in `RD_DATA` while `EMPTY` is low. The FIFO write pointer advances into new RAM locations (earlier results remain visible in the memory array) and `FULL` never asserts; `passed_tc` reaches 5 with `failed_tc` = 0.

TC	Start time	Trigger frames	Key evidence in waveform
1	≈ 643.7 µs	3 (RX)	`regArr[5]` = 0x55; no TX
2	≈ 1077.7 µs	2 (RX)	FIFO → `RD_DATA` 0x55 → 1 TX frame
3	≈ 1394.2 µs	4 (RX)	FIFO 0x25, 0x00 → 2 TX frames
4	≈ 2001.8 µs	2 (RX)	FIFO 0x1B, 0x00 → 2 TX frames
5	≈ 2418.5 µs	4 (RX)	FIFO 0x20, 0x01 → 2 TX frames

Table 9: Timeline of test cases (start times from the tc_start_time signal).

Figure 17 places the five test cases on the simulation time axis. After reset and the UART configuration writes, the cases run back to back and the last one finishes at about 3.10 ms.

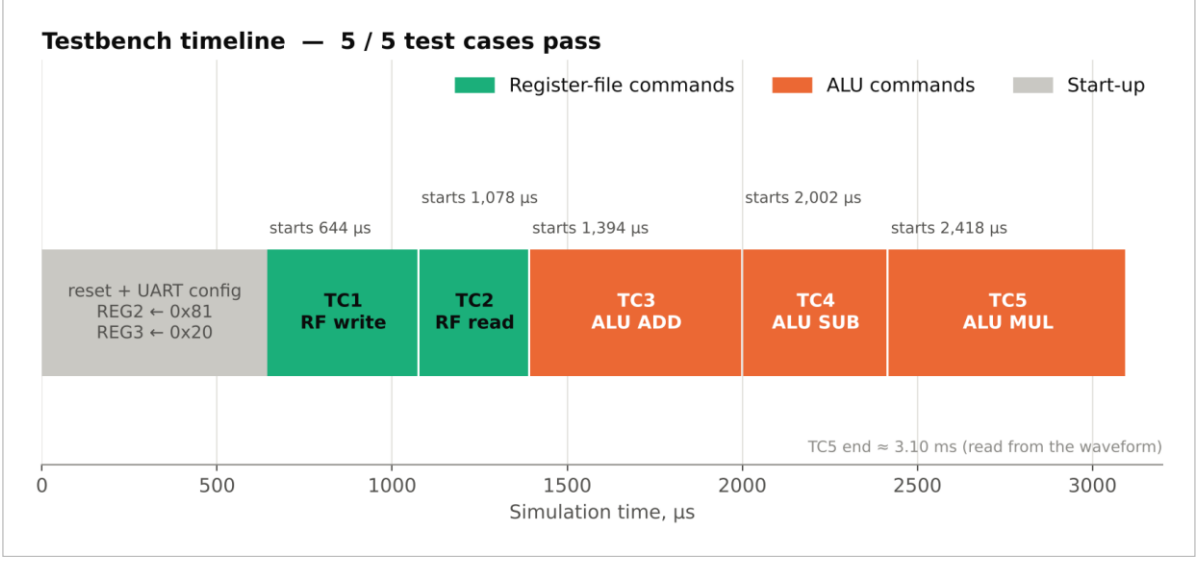


Figure 17: Testbench timeline, 5 / 5 test cases pass.

4.4 Synthesis Summary (Design Compiler)

Synthesis was run with Synopsys Design Compiler O-2018.06-SP1 using the TSMC 0.13 µm `scmetro_tsmc_cl013g_rvt` library at the slow corner (1.08 V, 125 °C). Hold analysis in the report uses the fast corner (1.32 V, −40 °C). The constraint report states: **"This design has no violated constraints."**

Metric	Value
Cells (total / combinational / sequential)	1945 / 1585 / 357
Buffers and inverters	259 cells (921.36 area units)
Combinational area	10970.37
Non-combinational (sequential) area	8947.63
Total cell area	**19918.00**
Hierarchical blocks kept	ClkDiv_RX 488.33 (2.5 %), ClkDiv_TX 643.65 (3.2 %), CLK_GATE 40.01 (0.2 %)
Total power (low-effort analysis)	0.227 mW (dynamic ≈ 0.217 mW, leakage ≈ 9.36 µW)
Constraint violations	None

Table 10: Synthesis area, power and constraint summary.

Path group	Check	Worst slack (ns)	Path	Result
ALU_CLK	Setup	10.06	RegFile `regArr[1][5]` → ALU `ALU_OUT_reg[0]` (arrival 9.39 ns)	MET
REF_CLK	Setup	14.92	SYS_CTRL `state_reg[2]` → RegFile `RdData_reg[0]`	MET
RX_CLK	Setup	215.83	RX stop-check → `RF_STP_ERR` output	MET
TX_CLK	Setup	6942.42	Serializer counter → `TX_OUT`	MET
UART_CLK	Setup	268.08	ClkDiv_TX counter [0] → [7]	MET
All groups	Hold	0.34 – 0.73	Tightest: RX data-sampling counter (0.34); loosest: SYS_CTRL → ALU `OUT_VALID_reg` (0.73)	MET

Table 11: Worst setup/hold slack per clock group (one worst path reported per group).

Figure 18 compares the worst setup slack of each clock on a logarithmic axis. The 50 MHz ALU and REF_CLK groups are the tightest; the slow UART-domain groups have slack of hundreds of nanoseconds or more.

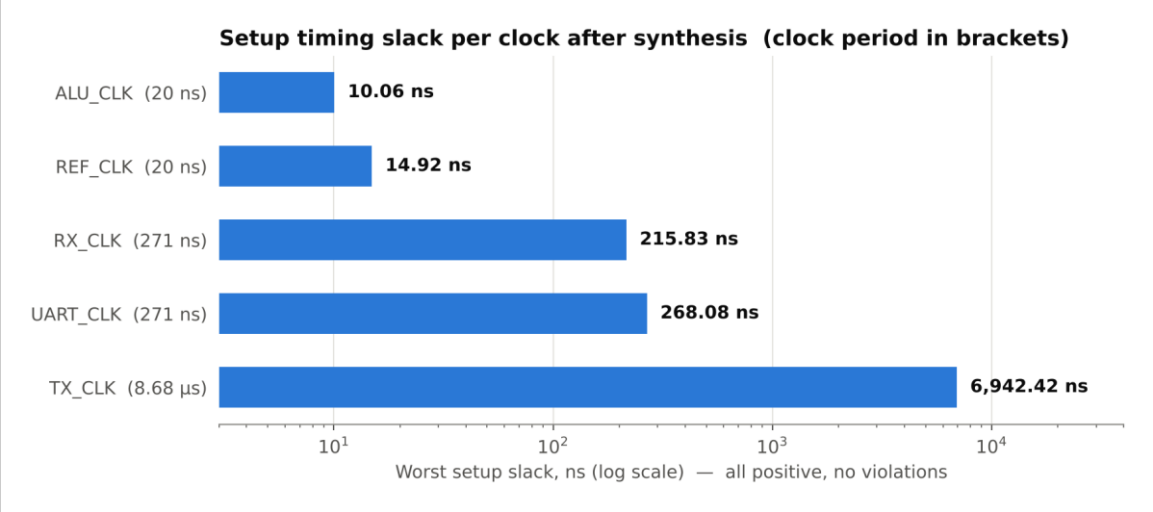


Figure 18: Worst setup slack per clock after synthesis (clock period in brackets).

Clock	Period (ns)	Source	Note
REF_CLK	20.00	Port	50 MHz
ALU_CLK	20.00	CLK_GATE/GATED_CLK	Generated, ÷1 of REF_CLK
UART_CLK	271.30	Port	3.6864 MHz
RX_CLK	271.30	ClkDiv_RX	Generated, ÷1 (prescale 32)
TX_CLK	8681.50	ClkDiv_TX	Generated, ÷32 → 115.2 kHz baud

Table 12: Clock definitions used in synthesis.

Log summary: compilation completed with no timing violations. The log contains 47 warnings, almost all lint-type notes: undriven-output cells (LINT-1) in ALU/ClkDiv/FIFO/UART blocks, constants on `U_ALU` and `Mux_4X1` pins (LINT-32), `clk_div_en` tied to logic 1 (LINT-52), and `CLKDIV_MUX` outputs [7:4] tied to 0 (LINT-31/52). These match the design intent (divider always on, prescale decoded to ≤ 4 bits).

4.5 DFT Summary (Scan Insertion)

Scan insertion used a multiplexed-flip-flop style with full-scan methodology and no clock mixing (`SCAN_CLK` 20 ns, `scan_rst` active-low, `test_mode` and scan-enable `SE` active-high). The script uses `-max_length 100`, so Design Compiler built **4 scan chains** from the 362 sequential cells (350 scan flops, 11 plain flops and 1 clock-gate latch). The top module declares `NUM_OF_CHAINS = 4`, giving scan ports `SI[3:0]` and `SO[3:0]`.

Chain	Scan ports	Cells	Starting element
1	SI[3] → SO[3]	91	RST_SYNC_1/Synchronizer_reg[0]
2	SI[2] → SO[2]	90	U_ASYNC_FIFO/FIFO_Memory_Block/RAM_reg[6][1]
3	SI[1] → SO[1]	90	U_RegFile/regArr_reg[3][1]
4	SI[0] → SO[0]	90	U_RegFile/regArr_reg[14][3]

Table 13: Scan chain architecture.

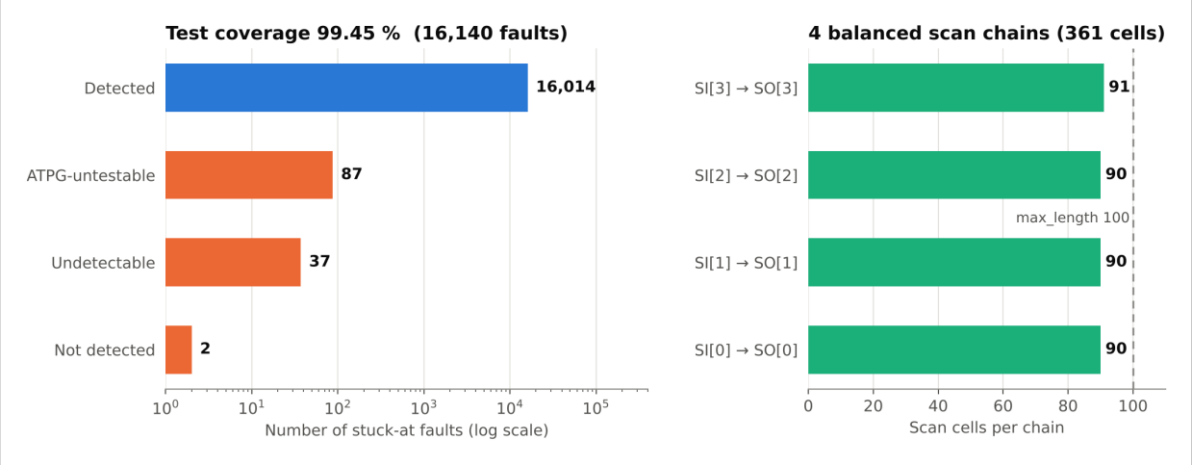


Figure 19: ATPG fault statistics and scan-chain balance.

DFT item	Result
Test design rule check	2 violations before insertion (TEST-126, TEST-505 on the clock-gate latch); 1 after insertion (TEST-505, expected: latch constant 1)
Sequential cells	362 (350 scan flops, 11 plain flops, 1 clock-gate latch); 361 cells in chains
Fault list (uncollapsed stuck-at)	16,140 faults
Detected / ATPG-untestable / undetectable / not detected	16,014 / 87 / 37 / 2
Test coverage	**99.45 %**

Table 14: DFT DRC and coverage summary.

Synthesis vs. post-DFT netlist

Metric	Synthesis	After DFT	Change
Total cell area	19918.00	22024.29	+10.57 %
Combinational area	10970.37	10571.47	−3.64 %
Sequential area	8947.63	11452.82	+28.0 %
Cells	1945	1845	−5.1 %
Total power (mW)	0.227	0.385	+70 % (approx.)
Setup, worst slack reported	10.06 ns (ALU_CLK)	13.01 ns (SCAN_CLK)	MET
Hold, worst slack reported	0.34 ns	0.32 ns (SCAN_CLK)	MET

Table 15: Effect of scan insertion. Sequential area grows because scan flip-flops include the input mux; combinational area shrinks as the original muxing is absorbed.

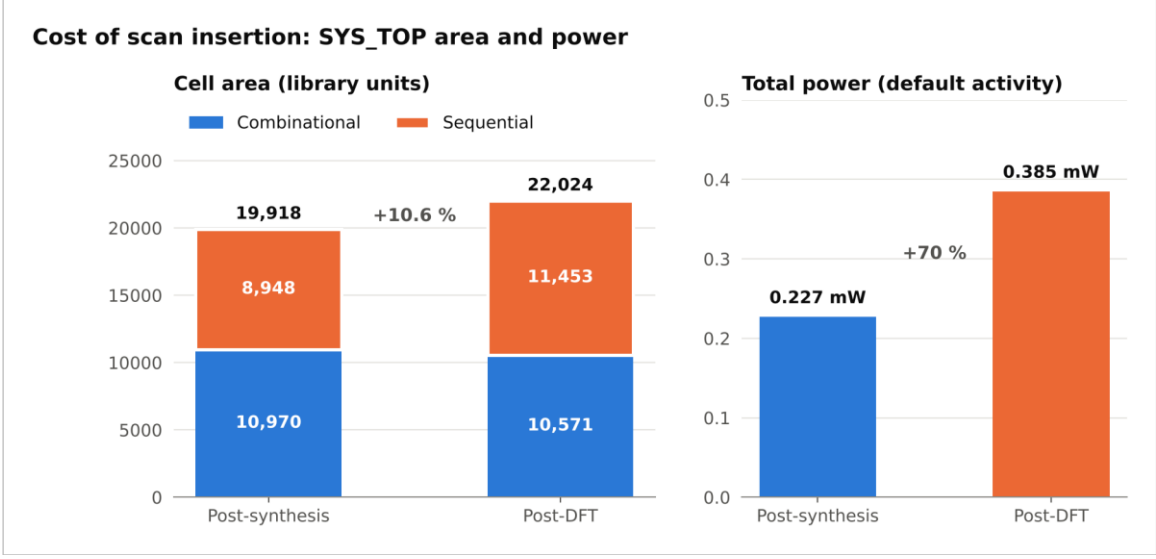


Figure 20: Area and power, post-synthesis vs post-DFT.

The added clock `SCAN_CLK` is defined at 20 ns alongside the functional clocks; all setup and hold checks in the DFT netlist are met.

After scan insertion most sequential cells are scan flops: 279 `SDFFRQX1M`, 64 `SDFFQX1M`, 4 `SDFFRQX2M`, 2 `SDFFSQX2M` and 1 `SDFFSQX1M`. 10 `DFFRQX1M` and 1 `DFFRQX2M` remain plain flops, and the clock-gate latch becomes a `TLATNX1M` cell (Figure 21).

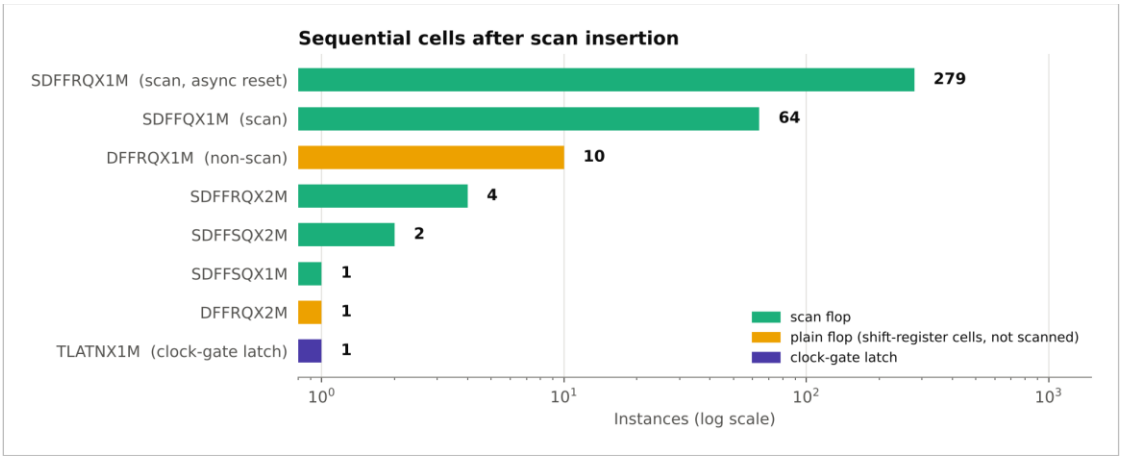


Figure 21: Sequential cells after scan insertion.

The DFT flow was repeated in `System_pnr/DFT` on `SYS_TOP_dft_pnr.v` to produce the netlist used for place and route (Section 4.8). That run also builds four chains of 91 / 90 / 90 / 90 cells and reports 16,124 faults with 99.48 % test coverage and 1,834 cells, and it writes the SDC files for the three analysis modes (func, scan, capture).

4.6 Formal Equivalence Summary (Formality)

Synopsys Formality L-2016.03-SP1 compared the RTL (reference) against each netlist (implementation), with `synopsys_auto_setup` enabled.

Comparison	Passing points	Failing / aborted / unverified	Verification	Elapsed
RTL vs post-synthesis netlist	358	0 / 0 / 0	SUCCEEDED	≈ 137 s
RTL vs post-DFT netlist	365	0 / 0 / 0	SUCCEEDED	—

Table 16: Formality results.

The post-synthesis comparison matched 358 compare points (3 ports, 354 flip-flops and 1 latch); the post-DFT comparison matched 365 (3 ports, 361 flip-flops and 1 latch). For the post-DFT run `test_mode` and `SE` were set to constant 0 (functional mode) and the eight scan ports `SI[*]` / `SO[*]` were excluded with `set_dont_verify_points`, so the scan-inserted netlist is functionally equivalent to the RTL in normal operation. There were 0 unmatched compare points and 6 unmatched unread points on the reference side. Formality noted that auto-setup mode was used and that RTL-interpretation messages occurred when linking the reference design (including an `FMR_ELAB-147` warning on the `Serializer` index), so results are valid under those settings. A post-place-and-route comparison has not been run: the `Backend/Formality/post-PnR` script is still an empty template.

4.7 SpyGlass Lint & CDC Summary

SpyGlass L-2016.06 was run in seven goals. Counts below are the "Results Summary" of each run; waived messages are listed separately.

Goal	Errors	Warnings	Info	Notes
Design_Read	0	0	7	SYS_TOP detected as the top unit; SDC → SGDC translation successful
lint_rtl	1	0	6	InferLatch in `CLK_GATE.v` line 24 (intended clock-gate latch); no combinational loops
cdc_setup_check	0 (1 waived)	0	14	Setup_port01 waived
cdc_clock_reset_integrity	0	0 (2 waived)	11	Clock_glitch04 ×2 waived
cdc_abstract	0 (1 waived)	0	12	Setup_port01 waived
cdc_verify_struct	0 (1 waived)	0 (1 waived)	40	Setup_port01, Ac_conv01 (+1 info) waived
cdc_verify	0 (1 waived)	0 (5 waived)	45	Full CDC verification — see below

Table 17: SpyGlass results by goal (reported messages after waivers).

CDC / reset metric (cdc_verify)	Result
Clocks / resets defined	2 clocks (REF_CLK, UART_CLK), 1 asynchronous reset
Registers / flat instances	367 / 1290
Unsynchronized scalar crossings (Ac_unsync01)	0 (7 synchronized)
Unsynchronized vector crossings (Ac_unsync02)	0 (14 synchronized)
Quasi-static crossings	15 (2 quasi-static constraints)
Reset synchronization	2 synchronized resets (RST_SYNC_1/2), both with synchronous de-assertion; 0 unsynchronized, 0 asynchronously de-asserted
Registers without clock / reset definition	0 / 0
Combinational loops	0
Black boxes	0

Table 18: CDC and reset-integrity summary.

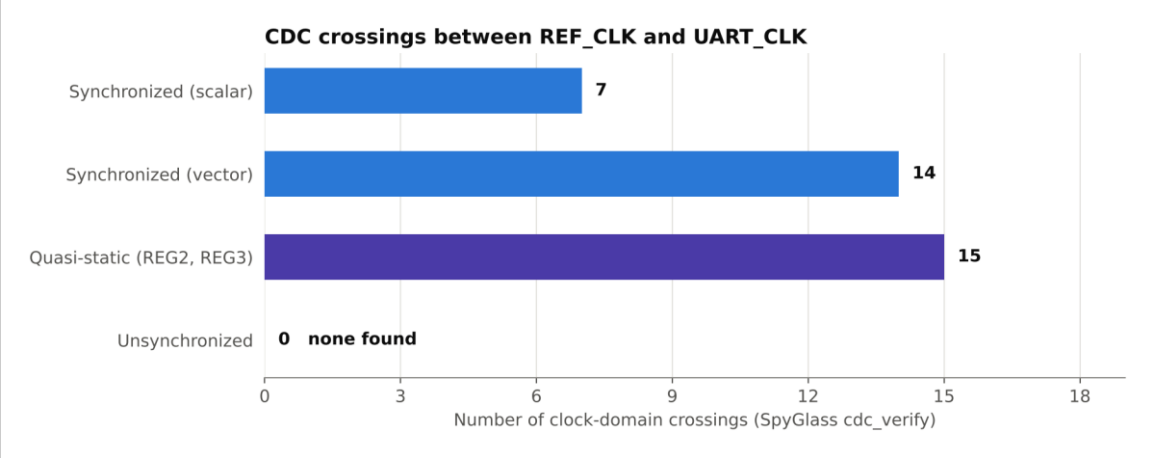


Figure 22: Clock-domain crossings between REF_CLK and UART_CLK, classified by SpyGlass.

Waived messages

Rule	Message (shortened)
Setup_port01	1 of 4 top-level ports (25 %) not associated with a clock constraint
Ac_clockperiod03	Design cycles between REF_CLK and UART_CLK exceed the analysis threshold
Ac_cdc01a	Fast-to-slow crossing from FIFO `wptr_inner[3]` into the `sync_w2r` synchronizer — data-hold check partially proved
Ac_conv01	5 synchronizers (FIFO r2w pointers and RX Data_Sync) converge on `SYS_CTRL.cmd_reg[7]`
Ac_datahold01a (×2)	FIFO RAM → UART_TX `parBit` and `pDataReg[7:0]`: data-enable sequencing partially proved
Clock_glitch04 (×2)	RegFile REG2/REG3 bits converge combinationally into the clock path of the UART FSM state flops

Table 19: Waived CDC/clock-reset messages.

4.8 Place and Route (Cadence First Encounter)

The scan-inserted netlist from the second DFT run (`SYS_TOP_dft_pnr.v`) and its SDC files were implemented in Cadence First Encounter 08.10-p004_1 with the TSMC 0.13 µm library

(scmetro_tsmc_cl013g_rvt, 7-metal technology LEF). The flow is split into one Tcl script per step (Figure 23).

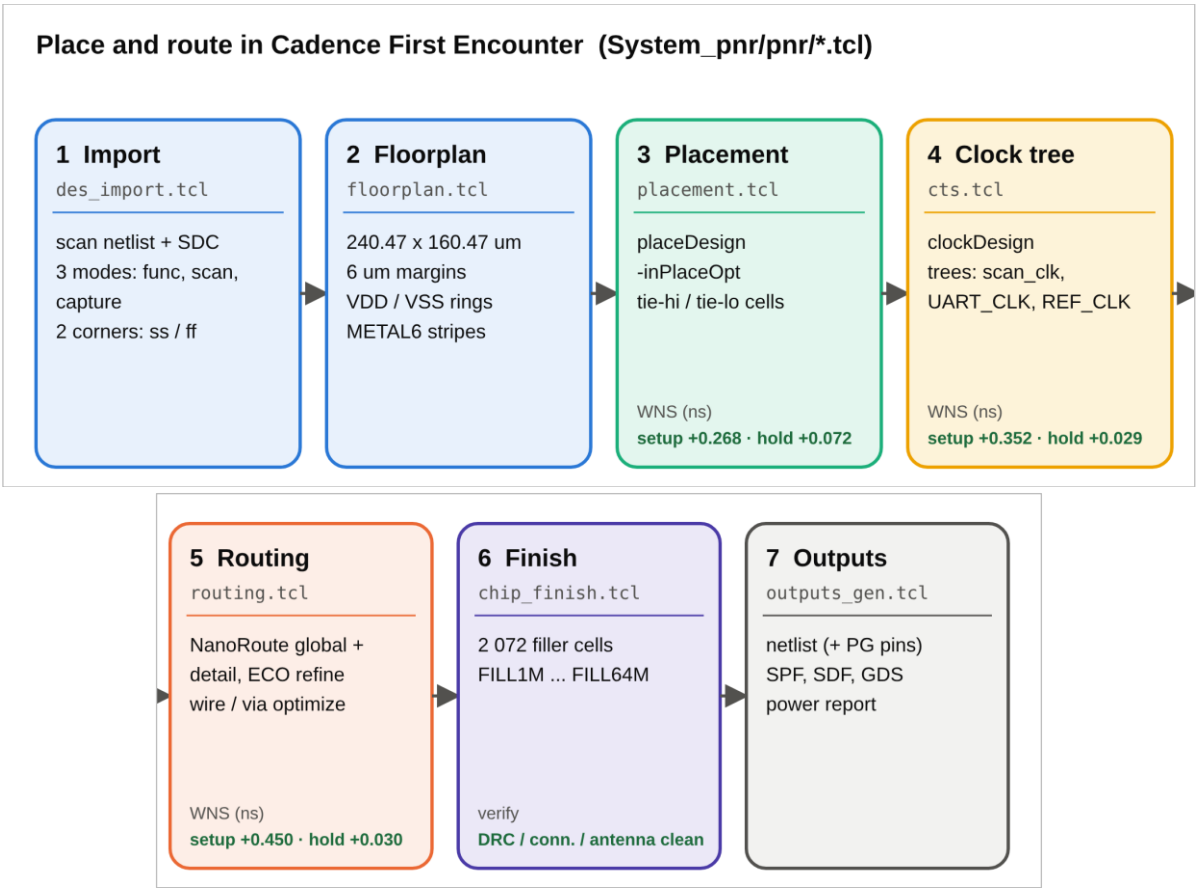


Figure 23: Place-and-route steps (steps 1–4 above, steps 5–7 below) with the result of each step; the sign-off results are in Table 23.

Setup

Item	Value
Analysis modes	func_mode (test_mode = 0, SE = 0), scan_mode (test_mode = 1, SE = 1), capture_mode (test_mode = 1, SE = 0)
Corners	max = ss 1.08 V 125 °C, min = ff 1.32 V −40 °C; one RC corner (tsmc13fsg.capTbl)
Analysis views	setup and hold for each of the three modes (6 views)
Die size	240.47 × 160.47 µm with 6 µm margins on every side
Power grid	VDD / VSS rings and METAL6 stripes (1 µm wide, 60 µm set-to-set)
Placement	placeDesign -inPlaceOpt -prePlaceOpt, tie-hi / tie-lo cells added
Clock tree	clockDesign with one tree each for scan_clk (root pin), UART_CLK and REF_CLK
Routing	NanoRoute global + detail routing with via / wire optimization and an ECO refine pass (refinePlace -preserveRouting)
Finish	2,072 filler cells (FILL1M … FILL64M)
Exports	netlist (with and without PG pins), SPF, SDF, GDS (≈ 199 KB), power report

Table 20: Place-and-route setup.

Timing

Stage	Setup WNS	Hold WNS	Violating paths	Placement density
Pre-CTS	+0.268 ns	+0.072 ns	0	66.7 %
Post-CTS	+0.352 ns	+0.029 ns	0	72.6 %
Post-route	+0.450 ns	+0.030 ns	0	100 % (fillers included)

Table 21: Worst negative slack (WNS) through place and route, all six analysis views.

After routing, setup WNS by path group is: reg2reg +0.450 ns, in2reg +15.559 ns, reg2out +13.147 ns and clock-gating +17.083 ns. There are no max-capacitance, max-transition or max-fanout violations (Figure 24).

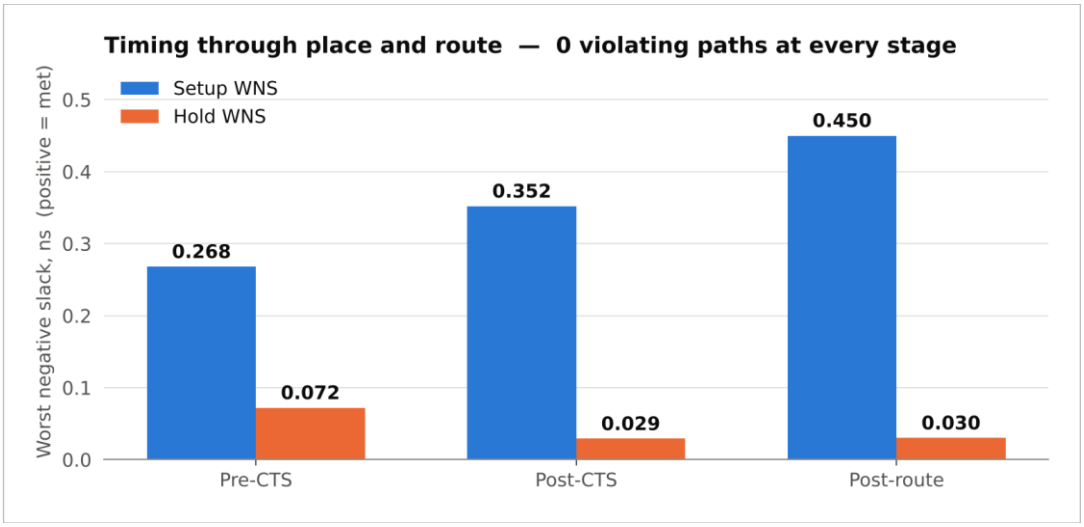


Figure 24: Worst slack at each place-and-route stage.

Clock trees

Tree	Buffers	Levels	Skew (setup views)	Skew (hold views)	Target
`scan_clk`	86	19	236.5 ps	96.8 ps	200 ps
`UART_CLK`	30	15	216.5 ps	–	200 ps
`REF_CLK`	42	11	92.7 ps	–	200 ps

Table 22: Clock-tree synthesis results.

The `scan_clk` and `UART_CLK` trees exceed the 200 ps skew target in the setup views, while `REF_CLK` is well inside it. Timing still closes in every view, so this is a quality-of-result note rather than a violation.

Sign-off checks

Check	Result
`verifyGeometry -noMinArea`	No DRC violations found
`verifyConnectivity -type all`	Found no problems or warnings
`verifyProcessAntenna`	No violations found
`timeDesign`	0 violating paths, 0 design-rule-violation nets

Table 23: Post-route sign-off checks.

Power

Post-route power (ss corner, 0.2 activity on primary inputs, no activity file) is **0.939 mW** in total: 0.628 mW internal, 0.296 mW switching and 0.016 mW leakage. Sequential cells account for 45.4 % and combinational cells for 54.6 %. This uses a different tool and activity assumption from the Design Compiler numbers in Section 4.4, so the two totals should not be compared directly. Figure 25 attributes the power to each clock.

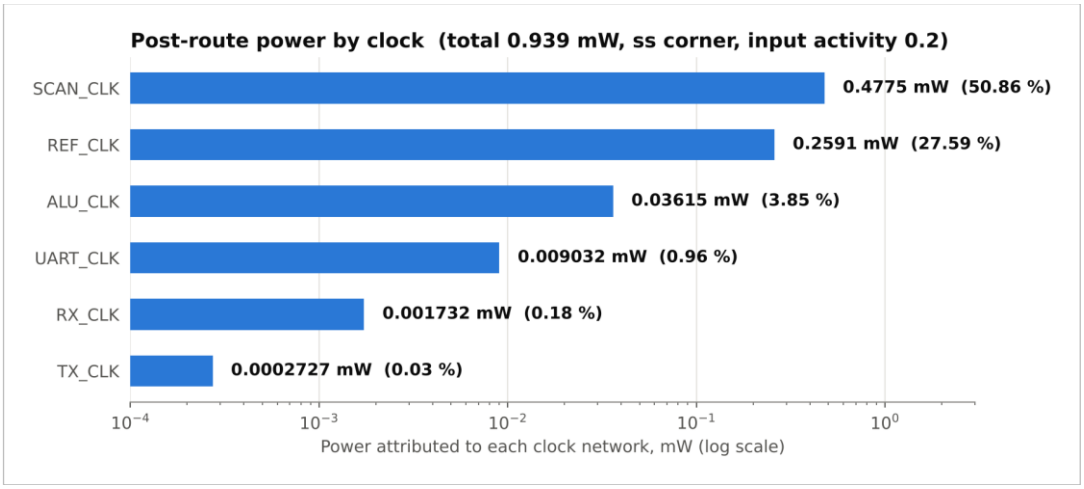


Figure 25: Post-route power attributed to each clock.

Layout

Figure 26 shows the final routed layout of SYS_TOP. Figure 27 and Figure 28 are Encounter screenshots from the floorplan, routing and module-placement views.

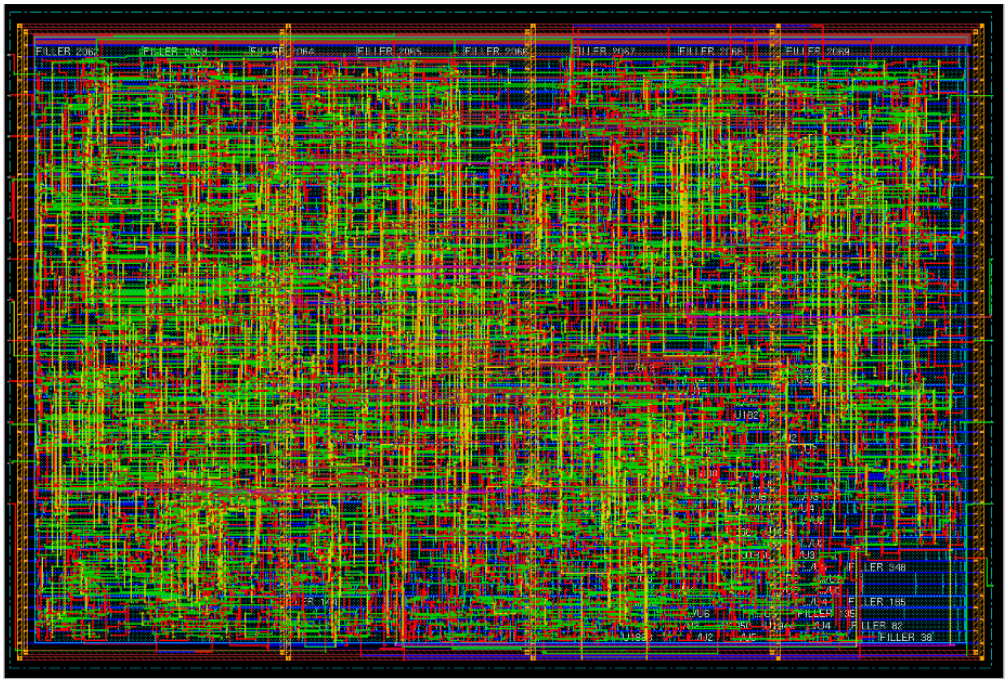


Figure 26: Routed layout of SYS_TOP (Results/Layout/Layout.png).

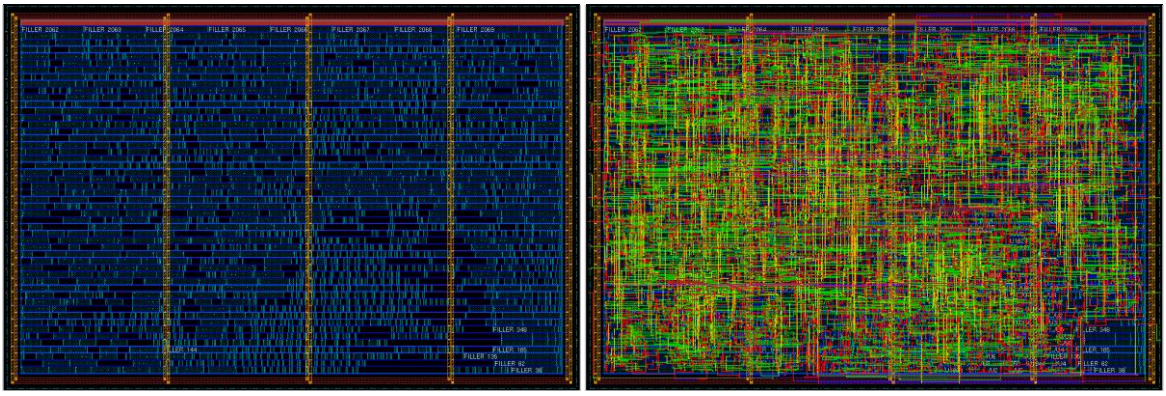


Figure 27: Encounter screenshots: floorplan with power stripes and cell rows (left) and routed design (right).

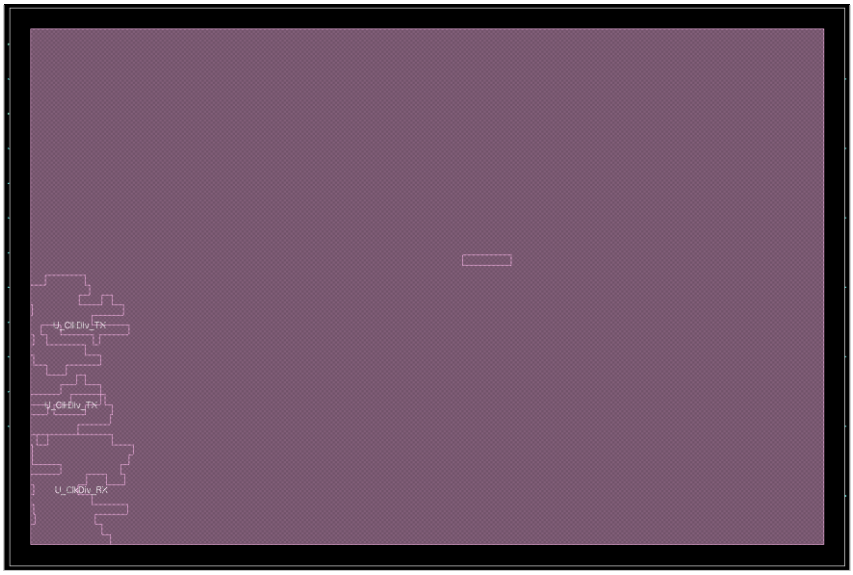


Figure 28: Amoeba (module-placement) view with the clock-divider modules U_ClkDiv_TX and U_ClkDiv_RX labelled.

05 DISCUSSION

Functional behaviour. All five test cases passed and the waveforms agree with the FSM design: frame counting in SYS_CTRL, FIFO hand-off to the TX domain, LSB-first transmission of the 16-bit ALU result, and operand reuse by the no-operand command. The 1.0 µs post-reset wait in the testbench is needed because the reset synchronizer takes about 0.54 µs (UART_CLK domain) to release.

Timing. Every path group meets setup with large margin after synthesis. The tightest is the ALU (10.06 ns slack on a 20 ns period, i.e. a 9.39 ns data path), so the 50 MHz target is conservative for this library corner; the UART-domain paths have slack of hundreds of nanoseconds because the clocks are slow. Hold slack is positive everywhere (≥ 0.32 ns). Those numbers use ideal clock networks. After place and route with real clock trees and wire delays, all six analysis views still meet setup (+0.450 ns WNS) and hold (+0.030 ns WNS), so the margin shrinks from nanoseconds to hundreds of picoseconds but stays positive.

Area and power trade-off. Scan insertion costs +10.57 % area (the flops become mux-D scan cells) and the reported power rises from 0.227 mW to 0.385 mW. In exchange the design reaches 99.45 % stuck-at test coverage. The power figures come from a low-effort analysis with default activity, and the scan netlist is analysed with the scan clock defined, so the comparison is indicative rather than a signoff number.

DFT limitations. The only DRC violation left after insertion is TEST-505 on the clock-gate latch (constant 1 in test mode); TEST-126 (latch not scannable) was reported before insertion. The residual 126 untested faults (87 ATPG-untestable, 37 undetectable, 2 not detected) are 0.78 % of the fault list. A natural follow-up is to check that the gated-clock ALU flops are reachable in shift/capture mode, for example by forcing the gate enable in test mode.

Formal equivalence. Formality proves the synthesized and scan-inserted netlists equal to the RTL (358 and 365 passing compare points, no failures). The scan-inserted result is valid for functional mode only (`test_mode` = 0, `SE` = 0), which is the standard setup. The routed netlist has not been compared against the RTL yet.

Lint and CDC. The single lint error is the latch inferred in the behavioural clock-gate model. It is intentional and is replaced by the library clock-gating cell during synthesis (Formality shows it as a latch compare point). CDC verification found **no unsynchronized crossings** (7 scalar and 14 vector crossings are all synchronized) and every reset is synchronized with synchronous de-assertion. The waived items are typical of an asynchronous-FIFO design — pointer and data-hold checks that SpyGlass can only partially prove — and static configuration registers that are written once at start-up. The waivers should be supported by a short justification in the project file; the unconstrained top-level port (Setup_port01) is worth closing by constraining it in the SGDC (the report does not name it here; the asynchronous `RX_IN` input is the usual candidate).

Place and route. The scan netlist was placed, clock-tree synthesized and routed with no timing violations and clean DRC, connectivity and antenna checks. The `scan_clk` (236.5 ps) and `UART_CLK` (216.5 ps) trees are above the 200 ps skew target in the setup views, while `REF_CLK` (92.7 ps) is inside it. Post-route power is 0.939 mW under the Encounter activity assumptions.

Remaining limitations. (1) The post-PnR Formality run is still an empty template. (2) Two DFT runs exist: `Backend/DFT` (99.45 % coverage) and `System_pnr/DFT` (99.48 %), and the place-and-route netlist comes from the second. (3) The clock-gate cell differs between netlists: the synthesis netlist uses the integrated `TLATNCAX12M` cell and the post-DFT netlist uses `TLATNX1M`. (4) The flow scripts use absolute paths from the original machine and must be edited before they are re-run.

Points to note

• The spec text gives the normal-access address range as starting at 0x4; with the default 4-bit address bus the register file has 16 locations (0x0–0xF).

• The ALU implements an extra A < B comparison (FUN = 1100) that is not in the spec list.

• Tools count registers differently (357 sequential cells in the synthesis area report, 362 in the post-DFT netlist and 364 in the DFT area report, 358 and 365 Formality compare points, 367 RTL registers in SpyGlass), so these totals are not expected to be identical.

• The report quotes the final `Backend/DFT` run for DFT results; the second DFT run in `System_pnr/DFT` feeds place and route and differs slightly (16,124 faults, 99.48 %).

06 CONCLUSION

- SYS_TOP was implemented per the specification, with two clock domains safely bridged by reset synchronizers, a data synchronizer and an 8-entry Gray-code asynchronous FIFO.
- The self-checking testbench passed **5/5** test cases covering register-file write/read and ALU add, subtract and multiply, and the waveforms confirm the expected frame-level behaviour.
- Synthesis produced a netlist of 19 918 area units with **no constraint violations** and positive setup/hold slack in all clock groups.
- Scan insertion added four chains for **99.45 %** test coverage at a +10.57 % area cost.
- Formality verified RTL ≡ post-synthesis and RTL ≡ post-DFT netlists (358 and 365 passing compare points, no failures).
- SpyGlass found no unsynchronized CDC paths; the only error is the intentional clock-gate latch, and a small set of waived warnings is documented above.
- Place and route in Cadence First Encounter closed timing in all six analysis views (post-route setup WNS **+0.450 ns**, hold WNS **+0.030 ns**) with clean DRC, connectivity and antenna checks and 0.939 mW post-route power.
- Open items: a post-layout Formality comparison, and reducing the `scan_clk` and `UART_CLK` clock-tree skew to the 200 ps target.